

AI Adoption Initiative Design Canvas

AI-Assisted Review for Senior Engineering Design Projects

Module 6 · Task 1 | Rey Granillo III | AI Campus Champions, Summer 2026

SECTION 1 — Problem Statement

Faculty and staff researchers who mentor senior engineering design teams must review lengthy technical reports on tight timelines across two semesters. This initiative uses AI to flag incomplete or problematic areas and generate targeted talking points, helping mentors deliver faster, more consistent feedback before Design Day.

SECTION 2 — Target Audience

Role(s)	Faculty and staff researchers serving as senior engineering design mentors
AI fluency (1–5)	2–3
What they need	A streamlined, semi-automated review process that directs them to the areas worth focusing on within long technical reports
Key concern	Credibility of AI-generated reviews. Addressed by keeping a human in the loop through chain-of-thought (CoT) prompting, in which intermediate natural-language reasoning steps make the model's review process transparent and checkable by the mentor before anything reaches a student.

Evidence: Wei et al. (2022) — chain-of-thought prompting elicits explicit reasoning in LLMs; Ji et al. (2023) — hallucination in generated text makes mentor verification a legitimate, necessary safeguard.

SECTION 3 — AI Tools and Approach

Tool	Modality	Rationale	Data Sensitivity	Approved?
Claude (via UA GenAI), using CoT	Text generation / reasoning	More accurate, meaningful output than alternatives for this review task; CoT surfaces reasoning for mentor review	Low — projects are built to be scaled and typically open-sourced	Yes
GitHub Copilot	Automated code review	Native to GitHub, where project code already resides; likely one- or zero-shot interactions	Low	Pending — Microsoft Copilot is on UA's Responsible AI list, but GitHub Copilot is not explicitly listed; to be confirmed

Evidence: Dell'Acqua et al. (2023) — within the 'jagged technological frontier,' AI meaningfully lifts knowledge-work speed and quality, supporting AI-assisted (not AI-replaced) report review.

SECTION 4 — Ethical Safeguards

Risk	Safeguard Mechanism	Responsible Party
Privacy violations — some projects may involve protected, regulated, or confidential data	Policy education on what is and is not safe to share with a model; restrict sensitive data from non-approved tools	Researchers and students providing the data
Intentional misuse	AI-use and disclosure policies; adherence to documented best practices	Model users (mentors and students)
Over-reliance / unverified output	Mandatory human-in-the-loop review of every AI-surfaced finding via CoT before it reaches a student	Mentors

Evidence: NIST AI RMF (2023) — grounds the govern/map/measure/manage approach to AI risk; Bender et al. (2021) — 'stochastic parrot' critique underlines why human oversight of model output is required.

SECTION 5 — Equity Considerations

Framed here as barrier-to-entry rather than the standard demographic equity dimensions.

Population	Barrier	Adjustment
Students	May struggle to use the AI tool effectively at key project checkpoints	Mentors, drawing on AI Champions training, provide guided direction on getting the most from the review system
Mentors	Vary in ability to engage an AI tool and steer it with CoT for complete, useful output	Share AI Champions course content and demonstrated examples of optimal AI engagement for reviews

Evidence: Wenger, McDermott & Snyder (2002) — knowledge stewarding within a community of practice frames mentors sharing AI fluency; Barocas, Hardt & Narayanan — fairness criteria inform the bias-minimizing persona in Section 7.

SECTION 6 — Success Metrics

Metric	How Measured	Timeline	Minimum Threshold
Student progress	Faculty-advisor check-ins across scheduled milestone presentations; graded critiques	Pre-set group presentation dates	All required points addressed at each presentation
End-of-project / final review	Final Design Day presentation + report, assessing whether AI-informed recommendations were correct / well implemented	End of two semesters	Passing grade — B or above

Evidence: Krathwohl (2002) / Anderson & Krathwohl (2001) — the revised Bloom's taxonomy grounds assessment of student progression from Apply toward Evaluate/Create across milestones.

SECTION 7 — Rollout: First Three Steps

Step	What Happens	Who / Resource	By When
1	Build an expert-reviewer persona using approved resources (UA GenAI or AWS Bedrock), with equity and ethics safeguards to minimize bias across project types and students	Me; UA GenAI / AWS Bedrock	Before fall semester begins
2	Provide relevant training data — aggregate past grades/scores, participating majors, presentation materials, and critiques from similar technology-focused projects to establish baseline model behavior	Me; historic project records	After the persona build
3	Testing — feed data as training/validation to sharpen recommendations (e.g., if a recommended focus area correlates with grade improvement, the model surfaces it when absent)	Me; project milestone data	Across fall & spring milestones

Evidence: Barocas, Hardt & Narayanan — independence/separation/sufficiency criteria inform the bias-minimizing persona design in Step 1; NIST AI RMF (2023) — iterative measure/manage cycle mirrors the Step 3 training-validation loop.

30-Second Elevator Pitch. Senior Design mentors face lengthy technical reports with limited time. Our initiative uses Claude's chain-of-thought reasoning to flag incomplete or problematic areas and generate targeted talking points—keeping faculty in control. The result: faster, more consistent, credible reviews that help engineering students strengthen their capstone projects before Design Day.

References

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